Fundamental Theorem of Calculus

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of fundamental theorem of calculus, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AC-02 / Advanced Calculus / Report v1

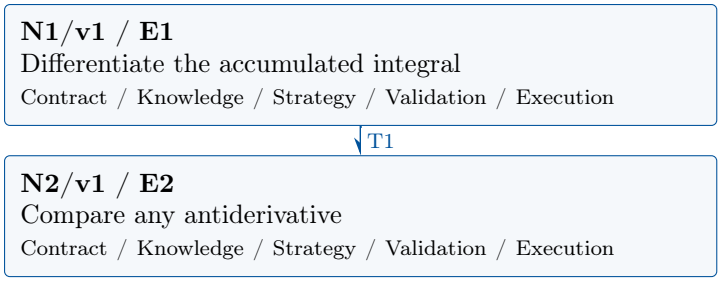
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For continuous $f : [a, b] \to \mathbb{R}$, prove $F(x) = \int_a^x f(t)\,dt$ is differentiable on (a, b) with $F' = f$. If $G' = f$ there and G is continuous on $[a, b]$, prove $\int_a^b f = G(b) - G(a)$.*

A. Foundations and permitted imports

Use Riemann integrability of continuous functions on[a,b], additivity and integral bounds, and AC-01. An integral with reversed endpoints changes sign. Derivatives of accumulated integrals are taken on (a,b), with one-sided endpoint continuity.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. F is continuous on[a,b] and differentiable inside with $F' = f$.

Receiving activity. N2: Compare any antiderivative

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: $G' = f$ inside and G continuous on[a,b]; prior exports E1/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Differentiate the accumulated integral

Object and role. Estimate the difference between a local average and $f(x)$.

Inputs. Continuous f; $F(x) = \int_a^x$ f; foundational imports stated in §1

C — Contract

Own E1/v1: F is continuous on[a,b] and differentiable inside with $F' = f$.

K — Knowledge

Retain Continuous f; $F(x) = \int_a^x$ f; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Estimate the difference between a local average and $f(x)$.

V — Validation

For negative h, the reversed interval must be treated consistently.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For x interior and sufficiently small nonzero h, additivity gives $(F(x+h) - F(x))/h = (1/h)\int_x^{x+h} f(t)dt$.

N1.2 Subtract $f(x)$; the absolute difference is bounded by sup between x andx+h $of|f(t) - f(x)|$, including $h < 0$ using reversed integration.

N1.3 Continuity makes this bound tend to 0.

N1.4 Thus $F'(x) = f(x)$.

N1.5 A global bound on|f| also $gives|F(x) - F(y)| \leq M|x - y|$, so F is continuous through the endpoints.

Export E1/v1. F is continuous on[a,b] and differentiable inside with $F' = f$.

Acceptance gate. For negative h, the reversed interval must be treated consistently. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: For negative h, the reversed interval must be treated consistently. Recheck dependent nodes N2.

B. N2/v1: Compare any antiderivative

Object and role. Use the mean value theorem on $G - F$.

Inputs. $G' = f$ inside and G continuous on[a,b]; prior exports E1/v1

C — Contract

Own E2/v1: The two asserted parts of the continuous-integrand FTC.

K — Knowledge

Retain $G' = f$ inside and G continuous on[a,b]; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use the mean value theorem on $G - F$.

V — Validation

An antiderivative must be continuous at the stated endpoints.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Let $H = G - F$.

N2.2 It is continuous on[a,b] and $H' = 0$ inside.

N2.3 For any $u < v$ in[a,b], AC-01 yields $H(v) - H(u) = H'(c)(v - u) = 0$.

N2.4 Thus H is constant.

N2.5 Since $F(a) = 0$ and $F(b) = \int_a^b$ f, subtract endpoint values to obtain $G(b) - G(a) = F(b) - F(a) = \int_a^b$ f.

N2.6 This closes the accumulation and antiderivative directions.

Export E2/v1. The two asserted parts of the continuous-integrand FTC.

Acceptance gate. An antiderivative must be continuous at the stated endpoints. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: An antiderivative must be continuous at the stated endpoints. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Differentiate the accumulated integral

1. For x interior and sufficiently small nonzero h, additivity gives $(F(x+h)-F(x))/h = (1/h)\int_x^{x+h} f(t)dt$.
2. Subtract $f(x)$; the absolute difference is bounded by sup between x andx+h $of|f(t) - f(x)|$, including $h<0$ using reversed integration.
3. Continuity makes this bound tend to 0.
4. Thus $F'(x) = f(x)$.
5. A global bound on|f| also $gives|F(x) - F(y)| \le M|x-y|$, so F is continuous through the endpoints.

B. §4.N2 — Compare any antiderivative

1. Let $H = G - F$.
2. It is continuous on[a,b] and $H' = 0$ inside.
3. For any $u < v$ in[a,b], AC-01 yields $H(v) - H(u) = H'(c)(v-u) = 0$.
4. Thus H is constant.
5. Since $F(a) = 0$ and $F(b) = \int_a^b$ f, subtract endpoint values to obtain $G(b) - G(a) = F(b) - F(a) = \int_a^b$ f.
6. This closes the accumulation and antiderivative directions.

□

V. WORKED ORCHESTRATION

For $f(t) = 3t^2$ on[0,2], $F(x) = x^3$ and $G(x) = x^3 + 7$. Then $G(2) - G(0) = 8 = \int_0^2 3t^2 dt$, while $F' = 3x^2$ demonstrates the local-average limit.

A. Boundary and hypothesis audit

A discontinuous integrand may be Riemann integrable without $F' = f$ at every point: take $f = 0$ except $f(0) = 1$ $on[-1,1]$. Its accumulated integral is 0, whose derivative at 0 is 0. Validation retains continuity for this pointwise FTC version.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

A discontinuous integrand may be Riemann integrable without $F' = f$ at every point: take $f = 0$ except $f(0) = 1$ $on [-1, 1]$. Its accumulated integral is 0, whose derivative at 0 is 0. Validation retains continuity for this pointwise FTC version.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.